Uncertainty and Sentiment in ARK Company Filings

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Repository: github.com/victor4628/FRE7871-NLP

I examine financial negative language and uncertainty in 1,624 eligible 10-K/10-Q filings from 91 ARK-held issuers, filed during 2021-2025. The clearest evidence is a rise in negative language within annual-report issuers. Uncertainty trends depend on report type and weighting. Its association with subsequent volatility weakens after controlling for existing volatility. Four-day return estimates are imprecise; they do not establish either predictability or the absence of an effect.

Table 1 Sample construction

Security and issuer screen	Removed	Remain
Unique cleaned securities in frozen snapshot	0	124
SEC ticker-to-CIK mapping found	7	117
At least one 2021-2025 10-K/10-Q	24	93
Unique companies (GOOG and GOOGL count as one)	1	92

Unique filing screen	Removed	Remain
Downloaded security-filing records	0	1,702
Duplicate Alphabet records (retain GOOGL)	20	1,682
Valid acceptance timestamp	0	1682
At least 2,000 narrative words	0	1682
Exclude explicitly identified shell companies	58	1624

Market-data requirement	Volatility: removed / kept	Returns: removed / kept
Starting unique filings	0 / 1624	0 / 1624
Complete 63-day pre-filing returns and positive volatility	32 / 1592	32 / 1592
Available dated share count and positive company market value	0 / 1592	0 / 1592
63 post-event daily returns available (volatility)	1 / 1591	Not applicable
4 event-day stock and SPY/ARKK returns available	Not applicable	0 / 1592

The initial 31 losses are 7 unresolved SEC ticker mappings and 24 mapped securities without an eligible 10-K/10-Q in the window. Filing counts for those excluded securities are not observed and are not invented. The 93 eligible securities represent 92 issuers because GOOG and GOOGL share filings. Explicitly reported shells are excluded. No minimum share-price threshold is imposed. Shell indicators and dated share counts are read from the corresponding reports. 15 otherwise eligible filings lack a usable share count; all 15 also lack the required prior-volatility history and therefore disappear at that earlier sequential screen. All download and parsing failures were zero.

Measures and identification

The instructor-specified LM lists contain 2,355 negative words and 297 uncertainty words. They overlap on 40 words, so the concepts are distinct but their empirical measures are not independent. Ten nonzero Negative flags actually mark removed entries; the required list is retained for comparability, with an active-only list as a sensitivity.

For term i in document j, I implement equation (1) as:
$\mathbf{w_{ij} = [(1 + \ln tf_{ij}) / (1 + \ln a_j)] \ln(N / df_i)}$ for tf > 0; otherwise zero.
Here a_j is total tokens divided by distinct tokens, not total document length. A category score sums its term weights. Proportions divide category occurrences by all tokens. IDF is recomputed on the exact document corpus used by each regression, including restricted sensitivities. This is full-corpus, retrospective scoring, not an out-of-sample forecast.

Parsing retains visible inline-XBRL text and removes hidden resources and mostly numeric tables (digit share above 15%). This extraction retains 62,590,430 tokens across unique downloads versus 48,537,683 under the starter parser, which also deleted visible tagged narrative. The starter uppercase tokenizer is retained; there is no stemming or stopword removal.

Table 2 Summary statistics by form

Form / measure	N	Mean	SD	P25	Median	P75
10-K / Negative %	396	2.347	0.406	2.084	2.361	2.620
10-K / Negative TF-IDF	396	163.684	54.480	122.192	155.045	204.915
10-K / Uncertainty %	396	1.947	0.240	1.803	1.969	2.119
10-K / Uncertainty TF-IDF	396	26.643	8.120	21.062	25.542	31.201
10-Q / Negative %	1,228	2.133	0.948	1.285	1.952	3.020
10-Q / Negative TF-IDF	1,228	80.297	66.244	24.021	56.244	131.109
10-Q / Uncertainty %	1,228	1.829	0.587	1.327	1.669	2.419
10-Q / Uncertainty TF-IDF	1,228	13.505	7.271	7.788	11.909	18.719

Negative-uncertainty correlations: 10-K prop: 0.722; 10-K tfidf: 0.838; 10-Q prop: 0.851; 10-Q tfidf: 0.921. Shared risk topics, overlapping vocabulary and document structure can make the scores move together. The separate outcome tests do not assume statistical independence.

Event timing and controls

Day 0 is the first exchange session closing strictly after the SEC acceptance timestamp, including early closes; 921 core filings move from their filing date. Four-day return is stock buy-and-hold return over [0,3] minus SPY buy-and-hold return. Volatility is daily-return SD times sqrt(252): [-63,-1] before and [4,66] after, excluding the four-day reaction. All 63 returns are required; missing prices are never filled.

Size is log market value: dated common shares from the same filing times day -1 nominal price, with split units aligned. It is an optional research control, included because company scale may affect both disclosure language and market behavior. Multiple common classes are summed and valued at the selected class price, an approximation. Weighted-average EPS shares are excluded. Tables 5-6 compare no controls, size only, prior volatility only and both on a common sample. The primary models contain no other controls or fixed effects. Prior return and turnover are not calculated. The 63 preceding returns provide prior volatility; size needs one preceding closing price and a share count. Standard errors cluster by issuer.

Reading the tables

The score row is the estimated association for a one-standard-deviation increase in the named language score. In Table 5 the score measures uncertainty; in Table 6 it measures negative language. Parentheses show the uncertainty of the estimated coefficient: standard errors allowing repeated reports from the same company to be related. The score p-value tests a zero score coefficient. The Holm-adjusted p-value accounts for multiple tests; Not applied means no adjustment was made for that column, not that its result is insignificant. Log means natural logarithm. SD means standard deviation; P25/P75 are the 25th/75th percentiles. The intercept is a fitted baseline and is included in every model.

Table 3 Words driving each measure

Shares below divide each word count by all occurrences of words on its own list, not by all words in the filings. The top 30 words account for 46.4% of Negative occurrences and 90.4% of Uncertainty occurrences. Uncertainty is particularly concentrated, which makes common-word downweighting consequential.

Rank	Negative word	List share	Uncertainty word	List share
1	LOSS	5.12%	MAY	35.22%
2	ADVERSELY	4.13%	COULD	18.54%
3	LOSSES	3.11%	RISK	4.98%
4	CLAIMS	2.97%	RISKS	4.48%
5	ADVERSE	2.77%	BELIEVE	2.93%
6	AGAINST	2.38%	APPROXIMATELY	2.56%
7	UNABLE	1.96%	ASSUMPTIONS	1.92%
8	LITIGATION	1.95%	INTANGIBLE	1.64%
9	HARM	1.87%	POSSIBLE	1.26%
10	FAILURE	1.81%	UNCERTAINTIES	1.16%
11	FAIL	1.34%	FLUCTUATIONS	1.12%
12	IMPAIRMENT	1.24%	MIGHT	1.06%
13	NEGATIVELY	1.14%	UNCERTAIN	1.02%
14	PENALTIES	1.14%	ANTICIPATED	0.99%
15	DIFFICULT	1.14%	PREDICT	0.95%
16	NEGATIVE	1.02%	VOLATILITY	0.94%
17	CRITICAL	1.01%	DEPEND	0.93%
18	DELAYS	0.94%	UNCERTAINTY	0.89%
19	DECLINE	0.92%	DIFFER	0.85%
20	DAMAGES	0.91%	ANTICIPATE	0.80%
21	RESTATED	0.85%	CONTINGENT	0.77%
22	LIMITATIONS	0.85%	EXPOSURE	0.77%
23	DELAY	0.84%	VARIABLE	0.69%
24	VOLATILITY	0.77%	DEPENDS	0.68%
25	CHALLENGES	0.76%	PENDING	0.65%
26	FINES	0.72%	DEPENDENT	0.58%
27	CLOSING	0.69%	CONTINGENCIES	0.55%
28	HARMED	0.69%	PROBABLE	0.51%
29	DISRUPTIONS	0.69%	ASSUMED	0.51%
30	INVESTIGATIONS	0.66%	VARY	0.48%

A frequently used word can dominate proportional counts yet convey little cross-document distinction. Equation (1) reduces that contribution through document frequency, while log term frequency dampens repetition. A high TF-IDF value can also reflect unusual vocabulary and document structure; it is not a probability of bad news.

Figure 1 Language scores and VIX

Each line first averages filings within issuer, form and filing quarter, then weights issuers equally. Annual and quarterly reports remain separate. VIX is the quarterly mean on the right axis; it is a market comparison, not the dependent variable in the firm-volatility regressions.

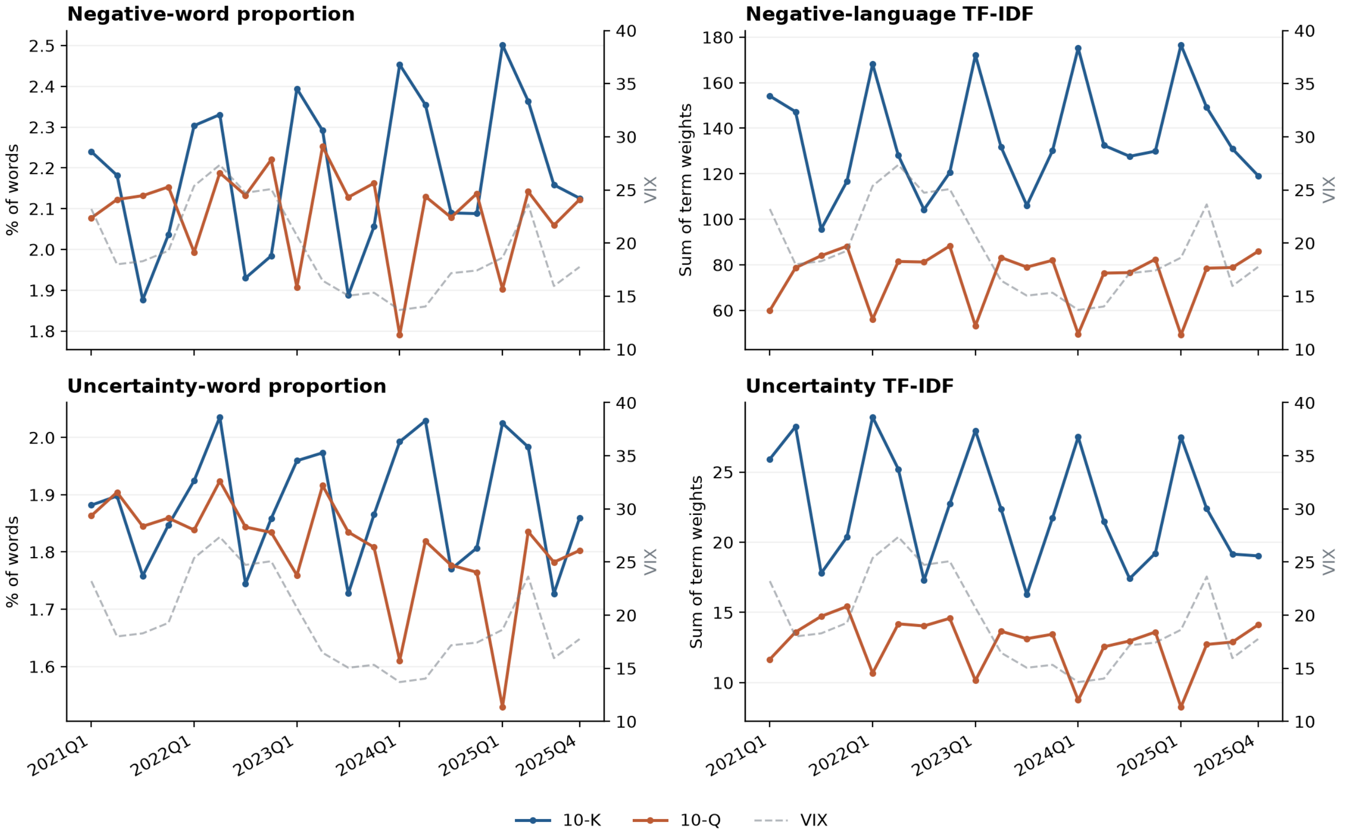

Table 4 Overall trends and changes within companies

Form	Measure	Overall slope	OLS t	NW t	Within-company slope	Within-company p	Adjusted p (Holm)
10-K	Negative %	0.0318	6.53	9.50	0.0345	<0.001	<0.001
10-K	Negative TF-IDF	0.0176	2.85	4.73	0.0133	0.002	0.007
10-K	Uncertainty %	0.0119	1.83	1.73	0.0249	<0.001	<0.001
10-K	Uncertainty TF-IDF	-0.0124	-1.44	-2.23	-0.0077	0.077	0.154
10-Q	Negative %	-0.0060	-2.57	-3.35	-0.0042	0.332	0.332
10-Q	Negative TF-IDF	-0.0062	-4.16	-4.06	-0.0072	0.113	0.154
10-Q	Uncertainty %	-0.0161	-4.30	-4.52	-0.0109	0.019	0.038
10-Q	Uncertainty TF-IDF	-0.0179	-6.28	-6.16	-0.0181	<0.001	0.001

Slopes are standard deviations per quarter. OLS means ordinary least squares; NW means Newey-West. Aggregate models include quarter-of-year effects, with Bartlett Newey-West SEs (4 lags, finite-sample correction) because quarterly residuals may be serially correlated; naive OLS t statistics are shown for comparison. Within-company models include issuer and seasonal effects and company-clustered SEs; issuers observed in only one quarter are omitted from that test. Holm adjustment covers the four within-company tests separately for each weighting scheme. Redundant fixed-effect columns are removed algebraically.

The within-company results, rather than the pooled chart, support the main trend interpretation. Annual-report negative language rises under both weights. Uncertainty does not have one universal trend: annual-report proportional uncertainty rises while its TF-IDF counterpart does not; quarterly-report uncertainty declines. Changing report mix and issuer mix therefore cannot be ignored. With only 20 aggregate quarters, even corrected aggregate inference is fragile.

Table 5 Uncertainty and subsequent volatility

The outcome is log annualized volatility on days [4,66]. Each panel compares exactly the same filings and standardized uncertainty scores under four control choices. No controls means the uncertainty score plus an intercept; size is log market value and volatility is log prior volatility. Parentheses contain company-clustered SEs.

Proportion

Variable / statistic	No controls	Size only	Prior volatility	Both controls
Uncertainty score (+1 standard deviation)	0.1204 (0.0368)	0.1170 (0.0261)	0.0257 (0.0102)	0.0381 (0.0089)
Company size (log market value)	-	-0.1237 (0.0131)	-	-0.0429 (0.0069)
Prior volatility (log)	-	-	0.7487 (0.0269)	0.6417 (0.0367)
Number of filings	1,591	1,591	1,591	1,591
Score p-value	0.002	<0.001	0.014	<0.001
Adjusted p-value (Holm)	Not applied	Not applied	Not applied	<0.001

TF-IDF

Variable / statistic	No controls	Size only	Prior volatility	Both controls
Uncertainty score (+1 standard deviation)	0.1382 (0.0267)	0.1064 (0.0234)	0.0418 (0.0096)	0.0435 (0.0092)
Company size (log market value)	-	-0.1188 (0.0138)	-	-0.0405 (0.0069)
Prior volatility (log)	-	-	0.7408 (0.0282)	0.6443 (0.0368)
Number of filings	1,591	1,591	1,591	1,591
Score p-value	<0.001	<0.001	<0.001	<0.001
Adjusted p-value (Holm)	Not applied	Not applied	Not applied	<0.001

Proportion: adding prior volatility changes the uncertainty-score coefficient from 0.1204 to 0.0257 without size, and from 0.1170 to 0.0381 holding size constant. TF-IDF: adding prior volatility changes the uncertainty-score coefficient from 0.1382 to 0.0418 without size, and from 0.1064 to 0.0435 holding size constant.

Adding size to the prior-volatility model raises the uncertainty coefficient. Conditional on prior volatility, uncertainty is positively associated with company size, while size has a negative coefficient in the full model. Omitting size therefore lowers the uncertainty coefficient. The change reflects a different conditional comparison, not stronger evidence of causation.

Prior volatility accounts for much of the pooled association. With both controls, the remaining association is significant after Holm adjustment across the four both-control outcome tests. This is a conditional association, not evidence that words cause volatility. The other columns show sensitivity to the controls; their p values are unadjusted.

Sensitivity of the model with both controls

Specification	Proportion: coefficient / p	TF-IDF: coefficient / p
Annual reports only	0.0566 / 0.009	0.0646 / <0.001
Quarterly reports only	0.0408 / <0.001	0.0417 / <0.001
Add company/time/type effects	-0.0088 / 0.393	-0.0047 / 0.692
Cluster by company and quarter	0.0381 / <0.001	0.0435 / 0.046

Form-specific samples re-estimate IDF. The firm FE sensitivity adds issuer, calendar-quarter and form effects; both uncertainty coefficients then become negative and insignificant. A robust within-company predictive relationship is therefore not established. Two-way clustering has only 20 time clusters; invalid nuisance variances are logged in the notebook. Annual- and quarterly-report estimates are positive, with larger point estimates for annual reports, but no interaction test establishes that the two forms differ.

Table 6 Sentiment and four-day excess returns

The outcome is stock minus SPY buy-and-hold return over [0,3], in percentage points. The same four models are estimated on a common sample within each weight. Parentheses contain company-clustered SEs; a dash means the regressor is omitted, not estimated to be zero.

Proportion

Variable / statistic	No controls	Size only	Prior volatility	Both controls
Negative-language score (+1 standard deviation)	-0.3546 (0.2808)	-0.3657 (0.2756)	-0.2614 (0.2848)	-0.2093 (0.2845)
Company size (log market value)	-	0.0498 (0.1172)	-	-0.1087 (0.1751)
Prior volatility (log)	-	-	-0.9208 (0.5999)	-1.1967 (0.8579)
Number of filings	1,592	1,592	1,592	1,592
Score p-value	0.210	0.188	0.361	0.464
Adjusted p-value (Holm)	Not applied	Not applied	Not applied	0.484

TF-IDF

Variable / statistic	No controls	Size only	Prior volatility	Both controls
Negative-language score (+1 standard deviation)	-0.5068 (0.2973)	-0.5043 (0.3063)	-0.3905 (0.3233)	-0.3781 (0.3209)
Company size (log market value)	-	0.0101 (0.1236)	-	-0.1258 (0.1742)
Prior volatility (log)	-	-	-0.8153 (0.6381)	-1.1191 (0.8644)
Number of filings	1,592	1,592	1,592	1,592
Score p-value	0.092	0.103	0.230	0.242
Adjusted p-value (Holm)	Not applied	Not applied	Not applied	0.484

All four specifications produce negative but statistically imprecise negative-language coefficients. With both controls, the estimates are -0.209 and -0.378 percentage points per SD. Failure to reject zero does not establish the absence of an economically meaningful effect.

Approximate 80%-power minimum detectable effects for the models with both controls are 0.80 and 0.91 percentage points per SD, using (cluster-t critical + 0.842) times the clustered SE. This is a precision diagnostic, not observed power. Holm adjustment covers only the four pooled both-control outcome tests.

Sensitivity of the model with both controls

Specification	Proportion: coefficient / p	TF-IDF: coefficient / p
Annual reports only	-0.495 / 0.270	-0.663 / 0.170
Quarterly reports only	-0.209 / 0.533	-0.455 / 0.265
Add company/time/type effects	-1.035 / 0.102	-1.157 / 0.156
Cluster by company and quarter	-0.209 / 0.476	-0.378 / 0.328
ARKK benchmark	-0.159 / 0.564	-0.129 / 0.701
Exclude retired Negative words	-0.202 / 0.478	-0.373 / 0.248

All return sensitivities remain statistically imprecise. In particular, replacing SPY with ARKK does not change the conclusion. Form-specific IDF is re-estimated on each restricted sample.

Limits and next step

The 124-security universe is selected from 2026 holdings snapshots, not historical ARK ownership. Only 78 of 91 retained issuers appear in all five filing years. The number of formerly held firms missing from this snapshot cannot be established from the supplied data. Survivorship can distort trends as well as return tests; fixed effects do not remove this selection. Other limits are 20 aggregate quarters, full-corpus rather than live scoring, noisy daily event timing, multi-class valuation proxies, and overlapping post-filing windows. A natural extension is a historical holdings universe with inclusion dates, followed by a held-out forecasting test. That change requires historical holdings data and rebuilding the sample according to each company's actual inclusion dates.

Sources: Loughran and McDonald (2011), Journal of Finance 66(1), 35-65, equation (1); Fall 2026 assignment brief; instructor starter repository; LM dictionary documentation; Yahoo price-adjustment documentation; exchange_calendars; statsmodels. AI assistance is disclosed in AI_USE.md; implementation choices beyond the brief are recorded in ANALYSIS_CHOICES.md.